



Phase 1 Storage Cost Review

DigitalOcean · 500 devices · \$10/mo renewal

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Platform planning · Version 1.1

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NYC1 Phase 1 topology · SaaS + MQTT + Archive + Managed Mongo

Executive Summary

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mooreVIEW Phase 1 — DigitalOcean Storage & Server Cost Review

500 devices · 50 MB/month per device · \$10/mo renewal · NYC1 Phase 1 topology

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Executive summary

Directional **cost and revenue** estimate for mooreVIEW **Phase 1 DigitalOcean** deployment (`do-inventory.md` / `CLOUD_DEPLOY_DO_PHASE1_ATL-MQTT.md`) at **500 monitored devices**, **50 MB/month** ingest per device, and **\$10/month** monitoring renewal per device.

Revenue @ \$10/mo per device

Metric	Monthly	Annual
Gross renewal revenue	\$5,000	\$60,000
Per device	\$10	\$120

Cloud infrastructure cost

Metric	Year 1 average	Year-end run rate
Data storage only	~\$180/year	~\$360/year
Per device (storage only)	~\$0.36/device/year	~\$0.72/device/year
All-in server stack	~\$1,119/year	~\$1,119/year
Per device (all-in server)	~\$2.24/device/year	~\$2.24/device/year

Margin (renewal revenue minus DO cloud stack only)

Metric	Monthly	Annual	% of revenue
Gross renewal revenue	\$5,000	\$60,000	100%
DO Phase 1 stack	~\$93	~\$1,119	1.9%
Gross margin after cloud infra	~\$4,907	~\$58,881	98.1%

Per device (annual): \$120 revenue – \$2.24 cloud cost = **~\$117.76 gross margin** (cloud infra only).

Not included in margin: hardware/commissioning, cellular (\$3/mo typical), DNS, backups, bandwidth overages, support labor, payment processing, or sales/marketing.

Revenue estimation @ \$10/mo per device

Assumes **500 billable devices** on renewal at list **\$10/month** (residential / light commercial band in Pricing Guide v2.4).

Line	Calculation	Monthly	Annual
Devices on renewal	—	500	500
Renewal rate	\$10/device/mo	\$10	\$120/yr
Gross renewal revenue	500 × \$10	\$5,000	\$60,000

Revenue vs cloud cost (per device)

	Per device/mo	Per device/yr
Renewal revenue	\$10.00	\$120.00
Cloud infra (all-in)	~\$0.19	~\$2.24
Cloud storage only (Y1 avg)	~\$0.03	~\$0.36
Cloud storage only (steady)	~\$0.06	~\$0.72
Gross margin after cloud infra	~\$9.81	~\$117.76

Break-even (cloud infra only)

Cost covered	Devices needed @ \$10/mo
Full Phase 1 stack (~\$93/mo)	~10 devices
Archive storage run-rate (~\$30/mo)	~3 devices

At 500 devices, cloud infrastructure is **~1.9% of renewal revenue** — storage alone is **~0.6%** at year-end run rate.

Reference configuration

Production inventory (NYC1 VPC):

Host	Role	Size
ccloud-1-saas-nyc1	SaaS nginx → :3100	4 GB / 2 vCPU / 80 GB SSD

cloud-mqtt-nyc	Mosquitto 1883/8883	2 GB / 1 vCPU / 50 GB SSD
archive-prod-nyc	Archive API :8090	2 GB / 1 vCPU + block volume
Managed MongoDB	Hot historian (7-day rolling)	10–20 GB tier

Data pipeline: appliances → MQTT → SaaS ingest → **Mongo hot (0–7 days)** → day-7 compaction → **zstd archive** on archive droplet volume.

Pricing basis (Aug 2026):

- Droplets: 4 GB = \$24/mo · 2 GB = \$12/mo
- Block storage (Volumes): **\$0.10/GiB/month**
- Managed MongoDB (smallest plan): **\$15.23/mo** (15 GB storage included); extra scalable storage **\$0.215/GiB/mo**

Data volume @ 500 devices × 50 MB/month

Layer	Calculation	Fleet total
Ingest rate	500 × 50 MB	25 GB/month
Hot Mongo (7-day rolling)	25 GB × (7 ÷ 30)	~5.8 GB steady state
Archive growth (Year 1)	25 GB × 12 months	~300 GB cumulative

The hot tier (~5.8 GB) fits within the **15 GB** base Managed Mongo plan — **\$0 marginal hot-storage cost** at this load.

The configured **200 GB** archive volume covers roughly **8 months** at this ingest rate; add ~~100 GB~~ (\$10/mo) before month 12.

Annual data storage cost (storage only)

Item	Year 1 (ramping)	Year-end run rate
Archive volume (300 GB × \$0.10 × 12)	~\$180 avg	\$360/year
Hot Mongo extra storage	\$0	\$0
Total storage	~\$180/year	~\$360/year

Per device (storage only):

- Year 1 average: **~\$0.36/device/year** (0.3% of \$120 renewal)
- Steady state (year-end): **~\$0.72/device/year** (0.6% of \$120 renewal)



2. At **~25 GB/month** ingest, SaaS and MQTT droplets likely remain at 4 GB / 2 GB for Year 1.
3. Mongo **compute** may need a step-up before **storage** if query/report load grows with fleet size.
4. Revenue scales linearly with devices (\$10/mo each); cloud stack is **largely fixed** until archive/compute tiers step up — margin improves with density until the next DO tier.
5. For comparison, Infrastructure Projections v1.0 assumes **~3 MB/day/site** compressed archive (site-level, not per-device).

Assumptions & disclaimer

Assumption	Value
Devices	500
Renewal rate	\$10/device/month (\$120/year)
Per-device ingest	50 MB/month (compressed archive equivalent)
Hot retention	7 days (Mongo), then zstd export
Archive retention	Cumulative (Year 1 model)
Region	NYC1 / NYC3 (DigitalOcean)
Topology	Phase 1 split: SaaS + MQTT + Archive + Managed Mongo

Disclaimer: Directional planning estimate for capacity, revenue, and capex review. Not a financial forecast, hardware quote, tax advice, or SLA commitment. DigitalOcean list prices as of document generation; formal quotes and renewal contracts may vary.

Phase 1 Storage Cost Review v1.1 — mooreVIEW platform planning.

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